scientific reports

OPEN

Data-driven EV charging infrastructure with uncertainty based on a spatial–temporal flow-driven (STFD) models considering batteries

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Electric vehicles (EVs) play a vital role in meeting the United Nations Sustainable Development Goals by 2030, primarily by reducing emissions and enhancing air quality. Properly positioning EV charging infrastructure within urban settings is essential to facilitate a shift toward sustainable transportation. As EV adoption grows, microgrids (MGs) face new challenges, including increased power losses, deterioration of voltage profiles, and voltage stability issues. Incorporating energy storage systems (ESS) can help address these challenges by improving overall system efficiency. This research proposes a comprehensive planning methodology that optimizes the placement of EV charging stations by considering traffic flow patterns over space and time. Additionally, a stochastic modeling approach is introduced to identify the optimal locations for ESS, taking into account uncertainties such as variable electrical loads and the intermittent nature of renewable energy sources (RESs). The ESS placement is modeled as a multi-objective optimization problem, aiming to enhance voltage stability, reduce power losses, and improve voltage profiles. The developed mathematical frameworks and algorithms facilitate the optimal sizing and siting of both EV charging stations and ESS. An economic evaluation of the MG, including the costs associated with ESS integration, is also incorporated. The effectiveness of this integrated planning approach is demonstrated through a case study on a representative transportation network, showing its capacity to mitigate the adverse impacts impacts of EV integration on microgrid performance. Furthermore, the study employs a stochastic framework to simulate and analyze the uncertainties inherent to electrical loads and renewable energy generation. The obtained results show that integrating ESS significantly improved voltage stability, with the minimum voltage stability index (VSI) value increasing from 0.5848 to 0.8631. It also reduced power losses by 33.34%, decreased transformer loading by 19.5%, and enhanced economic efficiency. Additionally, Sodium-Nickel Chloride (Na-NiCl2) offered the highest savings of 6.99%, demonstrating significant technical and financial benefits for the MG.

Keywords Charging infrastructure, Charging station, Energy storage systems, Smart cities, Renewable energy, Uncertainty, Sustainable Development Goals (SDGs)

The United Nations set two SDG for 2030, which are SDG number 7 (Affordable and Clean Energy) and SDG number 11 (Sustainable Cities and Communities)[1]. Plug-in EV are essential for attaining these goals by mitigating greenhouse gas emissions and enhancing air quality[2]. To foster the transition to clean energy transportation, it is crucial to position EV stations in cities strategically. Effective allocation of EV infrastructure necessitates a holistic approach that carefully considers both traffic flow and the impacts on the power distribution system[3]. Strategically placing charging stations along routes with high traffic flow ensures convenient access for EV users without causing disruptions or detours that might impede overall traffic movement. Simultaneously, understanding the implications on the power networks is crucial to prevent localized overloads and maintain

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grid reliability[4]. The anticipated power demand from charging facilities is expected to exceed the capacity of the power networks, consequently compromising the overall reliability of the power system[5,6].

Integrating ESS into the power network emerges as a viable solution to alleviate these adverse effects on the power system[7,8]. The effective integration of traffic optimization and power network considerations not only ensures a seamless and convenient user experience but also promotes cost-effective deployment by leveraging existing resources and minimizing unnecessary infrastructure investments. Therefore, carefully integrating these factors is essential for the sustainable and efficient growth of EV charging networks. Workplace chargers, public chargers, and large charging plants can all be used to charge these EV[4]. It is expected that this additional load, which quickly consumes a significant quantity of electricity from the power network, will negatively impact the power network if it is not controlled well and will increase the requirement for expensive infrastructure modernization, which will put a financial strain on utility companies[9]. Table 1 shows the challenges that should be confronted to intensify EV adoption in a power system.

Despite all these challenges, as fossil fuel supplies are limited and there are current environmental worries about greenhouse gas emissions, EV have gained popularity as viable transportation options in place of traditional automobiles[3]. Therefore, Plug-in EV have rapidly entered the electric power networks during the past ten years[10]. For this replacement to be successful, the difficulties and problems with EVs must first be resolved. EV charging from a power network with a large proportion of RESs reduces pollution emissions more[11,12]. As a result, MGs are being pushed as a way to address environmental and energy problems[13,14]. The absence of a charging infrastructure for electric cars is one of the most significant problems. Thus, the replacement of fuel-powered cars can be sped up by building and improving this infrastructure such that charging stations have excellent levels of accessibility in addition to other incentives[15,16]. Yet the exorbitant expense of constructing and outfitting charging stations makes meticulous planning even more important. To make the process of replacing these cars easier, this infrastructure must be properly and economically designed[17].

The source of electrical energy that these vehicles require presents another challenge when trying to charge them. Individual charging actions consume a significant amount of power quickly and discretely as an unusual sort of electric load[3]. Furthermore, as EV usage rises, more trustworthy EV charging stations with faster EV charging periods are required[18]. This leads to detrimental effects on an MG, including elevated voltage variations and power losses. Thus, in reality, the MG may suffer from high electrical power requirements brought on by the integration of EV[17]. The simultaneous consideration of transportation and power supply issues is required for fast charging site design. Ignoring any one of these factors might lead to bad financial decisions or even malfunctions in transportation or electrical systems[19]. For example, putting a rapid charging station close to the head of the feeder can help cut down on power supply losses. However, geographical constraints may make it more difficult for EV drivers to access this location[20,21].

An ESS is one of the promising facilities that can mitigate the challenges that result from the adoption of EV in an MG. ESS may improve the hosting capacity of the EV charging station to increase the number of EV that can charge from it simultaneously and boost MG's dependability to raise its resilience to recover swiftly from disruptions caused by the station[7,22]. MG gains a great deal from ES, including improved power quality and reliability, a higher share of RESs, a higher share of EV, mitigation of RESs intermittency-related problems, increased system stability, the ability to track loads and reduce peak loads, support for spinning reserve, and relief from transmission congestion[23–25].

In recent literature, many works have investigated EV charging to accomplish certain goals. The placement of the EV charging station has drawn the attention of certain studies to fulfill specific goal functions. However, determining the allocation of EV charging stations to achieve a certain goal function has piqued the curiosity of several studies. Table 2 shows an overview of recent research works that studied the EV charging stations whether its location only or location and size. Deb et al.[9] demonstrated how EV charging stations affect radial distribution network reliability indices, voltage stability, and power losses. Huang and Kockelman[10] introduced an optimization technique to obtain the optimal location of EV charging plants to maximize the profit considering the prices of installation, working, and land acquisition. An optimization approach was suggested by Ahmed and Weihua[15] to estimate the position of EV charging stations based on traffic flows, taking into account various types of charging services. Mao et al.[17]introduced a graph-based integrated placement planning approach that maximizes EV fast charging convenience while ensuring the dependable functioning of the electricity grid. Some researchers have been interested in determining the capacity and location of the EV charging station. Ahmet[7] suggested an optimization approach to obtain an optimal allocation of the EV , distributed generators,

Challenges	Description
Total power demand[3]	Total electrical load rises as the quantity of EVs on the road intensify
Transformer loading and aging[4]	Transformer overloading and aging result from high EV penetration
Power losses[4]	Power loss increases with a high quantity of uncontrolled EV charging stations
Cable loading[4]	Growth in EV penetration causes power lines and cables to become overloaded
Voltage stability[9]	Electrical power system stability decreases with a large quantity of unregulated EV charging stations
Voltage profile[10]	An increase in unregulated EV charging stations raises the possibility of voltage level-regulated limit breaches
Voltage and current unbalance[11]	Single-phase EV charging results in voltage and current imbalance due to various voltage dips between system phases
Harmonics distortion[11]	Harmonic distortion of voltage and current increases with high EV penetration

Table 1. Associated challenges in the adoption of electric vehicles (EV) in a power system.

Ref.	EV		Include RESs	ESS	Impact of EV			
	Loc.	Cap.	PV/WT	Loc./Cap.	P. Losses	Vol. Profile	Vol. Stability	Tr. Load
[9]	✓	–	–	–	✓	–	✓	–
[10]	✓	–	–	–	–	–	–	–
[15]	✓	–	–	–	–	–	–	–
[17]	✓	–	–	–	–	✓	–	✓
[19]	–	✓	–	–	✓	✓	–	–
[7]	✓	✓	✓	✓	✓	✓	–	–
[12]	✓	✓	✓	–	✓	✓	–	–
[20]	✓	✓	✓	–	✓	–	–	–
[18]	✓	✓	–	–	–	–	–	–
[26]	✓	✓	–	–	✓	✓	–	–
[27]	✓	✓	–	–	✓	✓	–	–
[28]	✓	✓	–	–	–	–	✓	–
[29]	✓	✓	✓	–	✓	✓	✓	–
[30]	✓	✓	–	–	✓	✓	–	–
[31]	✓	✓	–	–	✓	✓	–	–
[32]	✓	✓	–	–	✓	✓	✓	–
Proposed	✓	✓	✓	✓	✓	✓	✓	✓

Table 2. Overview of recent research on EV charging stations.

and ES to enhance the performance of power networks. To improve traffic and distribution networks, Sa'adati et al.[12] proposed an optimization technique to get the optimal allocation of EV fasting charging plants and RES, considering the deviation paths, the multiple shortest paths, and the uncertainty of RES. Amini et al.[20] introduced a two-stage optimization method that accounts for the distribution system's constraints as well as the overall income generated by EV to get the ideal allocation of the RES and EV fast charging station simultaneously. In the initial phase, parking lot investor and power system restrictions must be met to determine the best site for EV parking and RES. Reducing the overall power losses of the distribution power system is necessary for the second stage's best distribution of parking spaces for EV and RES.

Finding the allocation of the EV charging station while not considering RESs in the electrical power system has piqued the interest of certain researchers. Xiang et al.[18] introduced an innovative planning technique to get the optimal capacity and location of the EV charging plant, considering the destination traffic flow data. A meta-heuristic optimization technique was proposed by Chen et al.[26] to determine the best allocation of EV charging stations to reduce installation costs and maintain distribution system quality characteristics. Moradijoz et al.[27] presented a multi-objective optimization approach for obtaining the optimal EV parking allocation to maximize its revenue by using the EV as controllable loads. They operate in the off-peak time as electrical load and they operate in the peak time as supply considering the power system constraints. An approach was presented by Archana and Rajeev[28] to determine the best placement for EV charging stations while taking the distribution power system's power quality into account. based on harmonic distortion and reliability indicators. Zangmo et al.[29] explored the optimal placement of RDGs, including photovoltaic (PV), wind turbine (WT), and EVs, into Radial Distribution Systems (RDSs). This initiative aimed to minimize power loss and enhance both the voltage profile and stability index. To model solar irradiance and wind speed in specific regions, they utilized Beta and Weibull probability distribution functions. Additionally, electric vehicles were incorporated into the RDS framework using meta-heuristic algorithms to further reduce power loss and improve voltage stability. Prempeh et al.[30] investigated single-objective frameworks for the optimal simultaneous allocation of DG units and fast EV charging stations. They employed Particle Swarm Optimization and Artificial Bee Colony algorithms to perform the optimization tasks. The study analyzed the effects of optimal placement on the performance of the networks. Sadeghi et al.[31] suggested an optimal approach for the placement of EV charging stations, shunt capacitors, and DGs. Their strategy aimed to minimize losses and enhance voltage profiles through the reconfiguration and strategic allocation of these resources.Soliman et al.[32] presented an optimization framework to efficiently allocate and size fast charging stations, facilitate vehicle-to-grid integration, and optimize capacitor usage. The objective was to improve grid profitability by reducing capacitor costs and enhancing power quality metrics through a multi-objective function. The optimization process accounted for several constraints, including power balance, bus voltage levels, current flow limits, and specifications for capacitor banks.

Motivated by the emerging challenges, a framework is proposed to allocate EV charging infrastructure that maximizes the covered charging demand while minimizing the negative impact on the MG. This is achieved by considering the uncertainties in electrical load, WT, and PV systems when allocating ESS. The primary contributions of this research can be summarized as follows:

- A Spatial–Temporal Flow Capturing Location Model (ST-FCLM) is employed to strategically allocate EV charging stations within the transportation network. The ST-FCLM identifies optimal locations for different

types of charging facilities by analyzing the spatial-temporal distribution of traffic flows. Subsequently, queuing theory is applied to optimize the sizes of the allocated charging facilities.
- A data-driven method using a roulette wheel mechanism based on probability distribution functions is introduced to generate scenarios that model the variations in electrical demand and the intermittent output of WT and PV systems.
- The fast-forward scenario reduction technique is applied to decrease the number of scenarios, thus reducing the computational complexity of stochastic programming and transforming the problem into a more tractable form.
- The anticipated time-varying power demand from EV charging facilities is calculated. The subsequent negative impacts on the MG, such as power loss, voltage profile degradation, transformer loading, and voltage stability, are thoroughly analyzed.
- A novel mathematical model is developed to optimize the placement and sizing of the battery storage system (BSS) in the MG. This model, using the Symbiotic Organisms Search (SOS) Optimization, incorporates uncertainties in electrical load, WT, and PV systems. The multi-objective optimization minimizes total power losses, improves the voltage profile, and maximizes the voltage stability index.
- An economic analysis of the MG is included, taking into account the costs associated with ESS.

This paper is structured as follows: "Uncertainty characterization" section addresses the uncertainties related to electrical load, WT, and PV systems. Symbiotic organisms search" section explains the SOS Optimization technique. "Coupled power-transportation system modeling" section presents the integrated power-transportation system model. "Allocation framework for EV charging infrastructure and ES system" section outlines the proposed framework for allocating EV charging infrastructure and ESS. Finally, "Simulation results with analytical discussion" section presents simulation Results with analytical discussion to evaluate the proposed planning framework, and "Conclusions" section provides concluding remarks.

Uncertainty characterization

The scenario-based characterization of the uncertainty related to electrical load, WT, and PV is explained in this section.

Scenario generation

This section introduces a stochastic framework designed to effectively simulate the uncertainty inherent in electrical loads and RESs while maintaining suitable performance and minimal computational burden. The framework begins by considering a probability density function (PDF) for each unknown parameter. This initial step is crucial as it lays the foundation for modeling the variability associated with different factors. Next, the PDF is divided into several intervals, which helps to capture the nuances of the data more accurately[33]. A typical example of a PDF with seven distinct intervals can be seen in Fig. 1.

To generate a variety of scenarios with different probabilities, a roulette wheel mechanism is employed, as illustrated in Fig. 2. This mechanism allows for the creation of multiple scenarios, each reflecting a unique combination of parameters. For every scenario generated, the electrical load, as well as the power levels from PV and WT sources, can be expressed based on the probabilistic outcomes derived from the intervals. This methodology not only enhances the understanding of uncertainty in electrical systems but also provides a robust tool for analyzing the impacts of variability on system performance. By integrating these elements, the proposed framework offers valuable insights into energy management, supporting informed decision-making in the design and operation of energy systems[33].

For every scenario, the electrical load, PV and WT power levels may be stated as follows:

$$P_{sl-t}^{l} = P_{t}^{l,\ \mathrm{forc}} + \Delta P_{sl-t}^{l}\ t = 1, .., T; sl = 1, .., S_n, \tag{1}$$

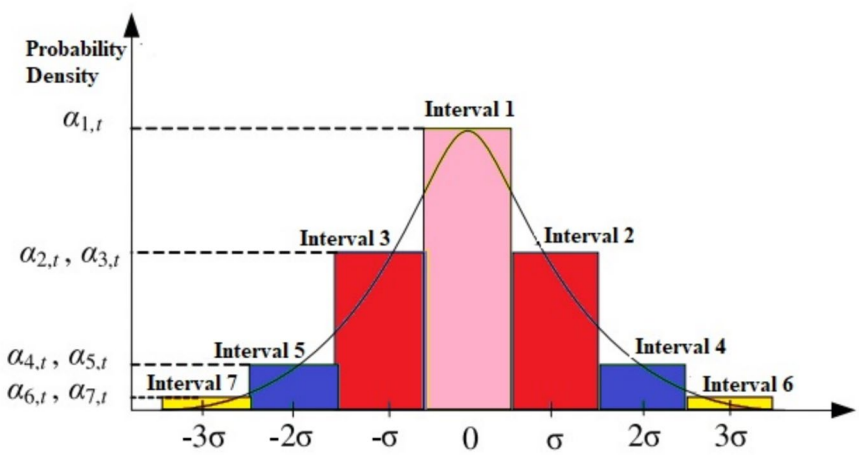

Fig. 1. Probability distribution functions of the load forecast error.

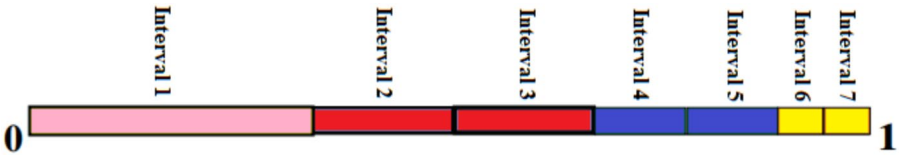

Fig. 2. Generating scenarios using the Roulette wheel mechanism.

$$P_{i,sp-t}^{pv} = P_{i-t}^{pv,\ \mathrm{forc}} + \Delta P_{i,sp-t}^{pv}, \quad i = 1,..,npv; t = 1,..,T; sp = 1,..,S_n \tag{2}$$

$$P_{i,sw-t}^{wt} = P_{i-t}^{wt,\ \mathrm{forc}} + \Delta P_{i,sw-t}^{wt}, \quad i = 1,..,nwt; t = 1,..,T; sw = 1,..,S_n \tag{3}$$

where P_{sl-t}^{l}, $P_{i,sp-t}^{pv}$, and $P_{i,sw-t}^{wt}$ are electrical demand in scenario *sl* at time *t* , output power of PV unit number *i* in scenario *sp* at time *t*, and output power of WT unit number *i* in scenario *sw* at time *t*, respectively. $P_{t}^{l,\ \mathrm{forc}}$, $P_{i-t}^{pv,\ \mathrm{forc}}$, and $P_{i-t}^{wt,\ \mathrm{forc}}$ are electrical load forecast value, forecast value of PV unit number *i*, and forecast value of WT unit number *i* at time *t*, respectively. ΔP_{sl-t}^{l}, $\Delta P_{i,sp-t}^{pv}$, and $\Delta P_{i,ws-t}^{wt}$ are forecast errors value of the electrical load in scenario *sl*, PVin scenario *sp*, and WT in scenario *sw* at time *t*, respectively. *T*, *npv*, *nwt*, and S_n represent the time interval index, PV units index, WT units index, and the scenario's total number. As seen in Fig. 1, each random variable's PDF is first divided into seven intervals centered on the zero means. The standard deviation of the electrical load or PV output power or WT output power forecast error (σ) determines the breadth of each interval. Additionally, there is a probability linked to each interval (*k*), represented by (α). The probability corresponding to each PDF's intervals is then normalized so that the aggregate of them equals unity. Furthermore, as seen in Fig. 2, every interval has a corresponding cumulative normalized probability.

Consequently, a vector of binary parameters designating the electrical load interval, PV output power interval, and WT output power interval in each period are included in each scenario:

$$\mathrm{scenario(s)} = \left[B_{(k,t,sl)}^{l},..,,B_{(7,t,sl)}^{l},B_{(1,i,t,sp)}^{pv},..,,B_{(7,i,t,sp)}^{pv},B_{(1,i,t,sw)}^{wt},..,,B_{(7,i,t,sw)}^{wt}\right]_{t=1,..,T}. \tag{4}$$

The binary parameters of intervals *k* at time *t* in the scenario for electrical load, output power of PV unit number *i*, and output power of WT unit number *i* are $B_{(k,t,sl)}$, $B_{(7,i,t,sp)}^{pv}$, and $B_{(7,i,t,sw)}^{wt}$, respectively.

Scenario reduction

The following formula, given in (5), is used to determine each generated scenario's normalized probability. $\alpha_{a,t}$, $\beta_{q,i,t}$, and $\gamma_{r,i,t}$ are probabilities of electrical load demand, PV output power, and WT output power for intervals a and q at time r, respectively. π_s is the probability of scenarios.

Indeed, modeling uncertainties with a large number of scenarios can significantly increase computational time and memory requirements. The fast-forward method is used in this paper to decrease the number of scenarios that are created[34]. Consider the original scenario set Λ with S_n scenarios and the chosen scenario set Λ_S and the final selected scenario set Λ_S^* with S_n^* scenarios. The algorithm's stages are as follows:

$$\pi_s = \frac{\prod_{t=1}^{24}\left(\sum_{a=1}^{7}\left(B_{a,t,sl}^{l}\times\alpha_{a,t}\right)\times\sum_{q=1}^{7}\left(B_{q,i,t,sp}^{pv}\times\beta_{q,i,t}\right)\times\sum_{r=1}^{7}\left(B_{r,i,t,sw}^{wt}\times\gamma_{r,i,t}\right)\right)}{\sum_{s=1}^{S_n}\left(\prod_{t=1}^{24}\left(\sum_{a=1}^{7}\left(B_{a,t,sl}^{l}\times\alpha_{a,t}\right)\times\sum_{q=1}^{7}\left(B_{q,i,t,sp}^{pv}\times\beta_{q,i,t}\right)\times\sum_{r=1}^{7}\left(B_{r,i,t,sw}^{wt}\times\gamma_{r,i,t}\right)\right)\right)}. \tag{5}$$

Step 1: Determine the separation between every pair of scenarios by using:

$$v\left(\Upsilon,\Upsilon'\right) = \sum_{\mathrm{load,\ PV,\ \&\ WT}}\left|P^{\Upsilon}-P^{\Upsilon'}\right|. \tag{6}$$

Step 2:Determine the average separation between every scenario and the rest. In this stage, the scenario with the smallest distance is chosen as the initial scenario, and the non-selected scenario set Λ_J and the selected scenario set Λ_S will be updated:

$$\Upsilon_1 = \arg\left\{\min_{\Upsilon'\in\Lambda}\sum_{\Upsilon\in\Lambda}\pi_{\Upsilon}v\left(\Upsilon,\Upsilon'\right)\right\}, \tag{7}$$

$$\Lambda_S = \Lambda_S\cup\left\{\Upsilon_1\right\}, \tag{8}$$

$$\Lambda_J = \Lambda_J\backslash\left\{\Upsilon_1\right\}. \tag{9}$$

Step 3: Determine the separation between every non-selected scenario and the ones that were chosen. The following scenario is chosen, and both the selected and non-selected scenario sets will be changed, based on which scenario minimizes this distance:

$$\Upsilon_i = \arg\left\{ \min_{\Upsilon' \in \Lambda_J} \sum_{\Upsilon \in \Lambda_J \setminus \{\Upsilon'\}} \pi_\Upsilon \min_{\Upsilon'' \in \Lambda_S \cup \{\Upsilon\}} v\left(\Upsilon, \Upsilon''\right) \right\}, \tag{10}$$

$$\Lambda_S = \Lambda_S \cup \{\Upsilon_i\}, \tag{11}$$

$$\Lambda_J = \Lambda_J \setminus \{\Upsilon_i\}. \tag{12}$$

Last step: The chance of each non-chosen scenario is added to the likelihood of the nearest picked scenario in the final step:

$$\pi^*_\Upsilon = \pi_\Upsilon + \sum_{\Upsilon' \in J(\Upsilon)} \pi_{\Upsilon'}. \tag{13}$$

$J(\Upsilon)$ is the non-selected scenario set that most closely resembles the chosen scenario Υ:

$$\Upsilon = \arg\left\{ \min_{\Upsilon'' \in \Lambda_S} v\left(\Upsilon'', \Upsilon'\right) \right\}; \Upsilon' \in \Lambda_J. \tag{14}$$

Coupled power-transportation system modeling

Consider a metropolitan area where the MG and the Road Transportation Network (RTN) are connected using geographic data. The deployment of EV charging infrastructure is planned within this urban setting.

EV charging infrastructure

The collection of charging stations for electric vehicles is represented by the notation $\mathrm{N} = \{1, 2, \ldots, N\}$. There are three subsets of this infrastructure: Subset F comprises DC fast charging stations (FCSs), subset O comprises on-road wireless chargers (OWC), and subset P comprises parking lots (PL) with AC chargers. EV users have the flexibility to choose among these options. Each EV initiates its journey with a minimum half-full battery charge. In cases where the travel distance surpasses the EV's range, the initial charge assumption dictates the required number of charging facilities between origin-destination pairs.

Facilitating charging involves connecting a cable to a charger in PL or FCS. It's worth mentioning that charging times in PL are relatively longer compared to those in FCS. As a result, FCS are strategically deployed at key transportation system nodes like road crossings, whereas PL are carefully positioned at destination nodes for everyday journeys like shopping malls and places of employment. Some EV use wireless power transfer in OWCs on designated lanes. These charging lanes allow EV to enter, charge wireless, and then easily return to regular traffic flow.

Road transportation network

In a typical urban area, an urban RTN comprises nodes N^T and directed links L^T connecting intersections, highway exits, or high-traffic locations. Origins $\mathscr{O}$ and destinations $\mathscr{D}$ represent where vehicles start and end daily trips, with traffic demand at origins being deterministic and independent of network conditions. At first, the RTN is empty ($t = 0$). It should be noted that User Equilibrium is defined by individual drivers searching for routes that reduce trip times[35].

Microgrid configuration

The modified IEEE 33 bus system developed in Ref.[23] is used in the following to verify the proposed methodology in this study. Five PV units, five WT, and an MG central controller (MGCC) make up the typical MG under consideration, as seen in Fig. 3. To improve the performance of the MG, MGCC is in charge of efficiently managing the energy from the utility, WT, and PV in addition to the charging and discharging of ESS. The information for each PV unit and WT is taken from Ref.[23] and is shown in Table 3.

The uncertainty surrounding the creation of RESs is one of the main obstacles. However, there is also a degree of uncertainty surrounding the electrical load. To address the uncertainties associated with load and RESs, this research offers an uncertainty modeling method using a roulette wheel mechanism that is based on the PDF to handle the uncertainties related to electrical load and RESs.

Using historical data, a thousand scenarios have been created to predict electrical load, PV, and WT uncertainty. The number of possible outcomes for electrical load, PV, and WT is then decreased using the fast-forward selection scenario reduction technique to ten scenarios. Figures 4, 5, and 6 show the clustering scenarios of the hourly electrical load, the hourly PV output power, and the hourly WT output power, respectively. In addition, Figs. 4, 5, and 6 show the constant hourly electrical load, the constant hourly PV output power, and the constant hourly WT output power, respectively, without considering the load uncertainty.

Figure 7 shows the probability of the reduction scenarios (ten scenarios) based on their perceived likelihood or frequency of occurrence using the Fast-Forward Selection technique.

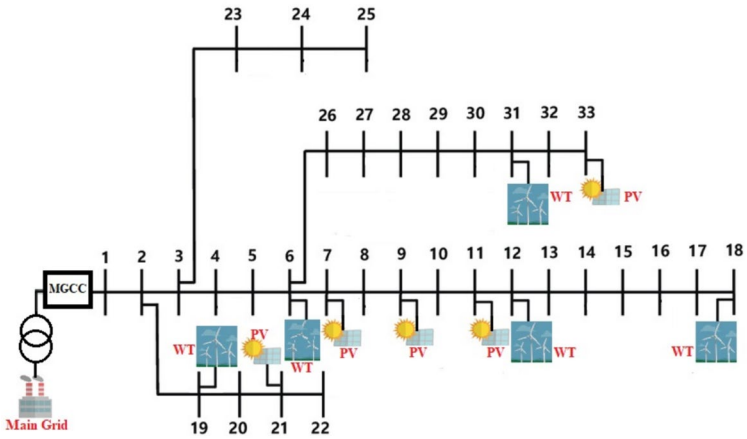

Fig. 3. The microgrid structure.

PV	Location	7	9	11	33	21
	Size (MW)	0.24	0.36	0.36	0.6	0.36
WT	Location	6	12	19	31	18
	Size (MW)	1.2	0.6	0.96	1.2	0.6

Table 3. The location and size of WT and PV units.

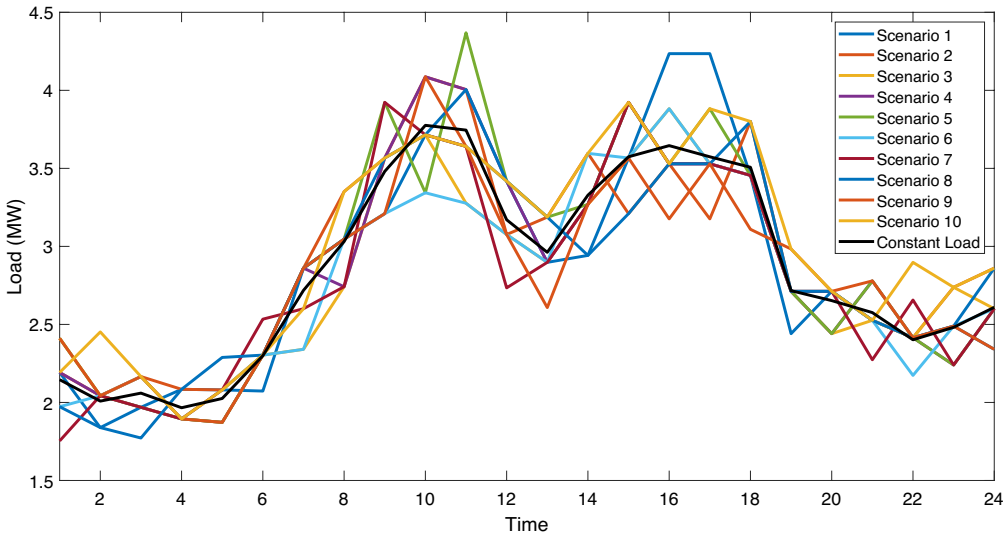

Fig. 4. Clustering scenarios of the hourly MG electrical load.

Allocation framework for EV charging infrastructure and ES system

EV charging facility allocation

we employ a sophisticated model called the ST-FCLM that was created to situate various EV charging stations on the transportation network[15].

The ST-FCLM plans three facility types—OWCs, FCS, and PL—addressing the diverse needs of EV users. Whether charging *en route* between origins and destinations or statically at trip endpoints, users seek flexibility. EV drivers typically charge at trip destinations if their EV have sufficient charge to complete the trip, avoiding wait times. *En route* charging occurs when battery state-of-charge (SoC) is low or for long-distance trips. OWCs and FCS are ideal for *en route* charging, as users prefer shorter charging times during trips. OWCs, strategically located to capture peak traffic during congested periods, enable efficient charging due to slower speeds. Meanwhile, FCS address *en route* charging during non-peak hours, as users are more likely to spend time at these facilities. For *static* charging at trip destinations during extended parking periods, PL are essential. This comprehensive approach ensures that *en route* charging is addressed by OWCs and FCS during travel, while PL

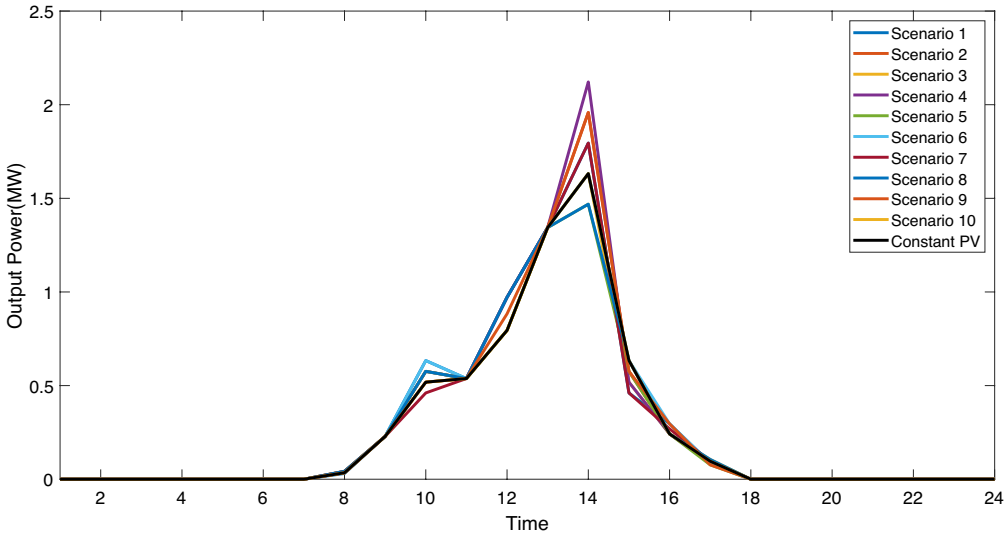

Fig. 5. The clustered hourly PV output power scenarios.

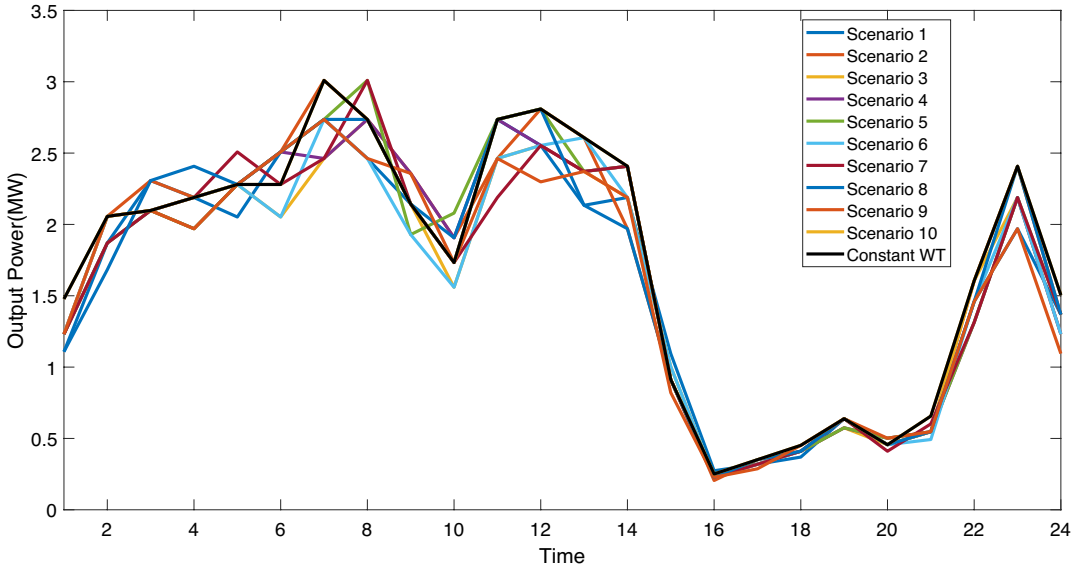

Fig. 6. The clustered hourly WT output power scenarios.

handle *static* charging during parking as shown in Figure 8. To optimize siting, the ST-FCLM model takes into account the distinct qualities and consumption patterns of each type of facility.

Using a dynamic traffic assignment model, the ST-FCLM estimates time-varying traffic flows on RTN. Following this, it employs the Gaussian mixture model to categorize the traffic flow dataset into distinct clusters. Every kind of charging station is arranged to correspond with certain traffic patterns. Finding the best sites to optimize the collected traffic flows for each kind of charging facility is the main objective of the ST-FCLM. In Ref.[15], the ST-FCLM's mathematical formulation is described in full.

After determining optimal locations for each charging facility type, the subsequent step involves refining their sizes through queuing theory. The goal is to optimize the number of chargers in each facility, ensuring they meet specific quality-of-service (QoS) targets, particularly concerning the average waiting time for customers during peak traffic hours[36]. This optimization process operates under several key assumptions. Firstly, EV arrivals at charging facilities are modeled as a Poisson process, where the mean arrival rate (λ) is directly proportional to the captured traffic volumes and the EV penetration rate. In this study, arrival rates at charging facilities are estimated using outputs from a road traffic simulation model. Furthermore, assuming constant charging power, the charging duration of EV is modeled as an independent, exponentially distributed variable. The mean charging duration is $1/\mu$, where μ is the service rate representing the number of EV that can be charged per unit time, as proposed in the EV charging behaviour model in Ref.[37].

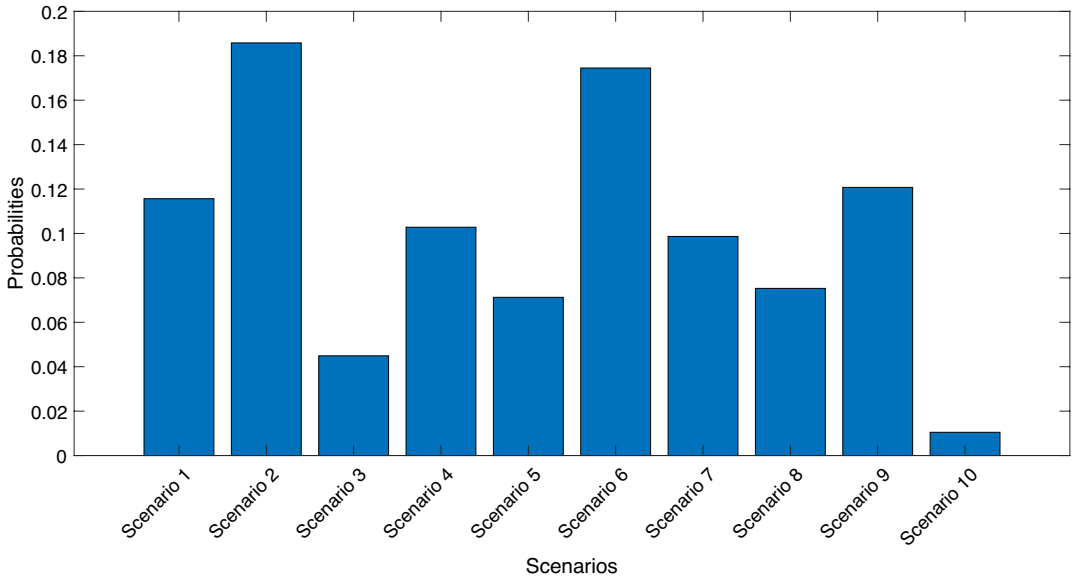

Fig. 7. Probability of occurrence of the reduction scenarios.

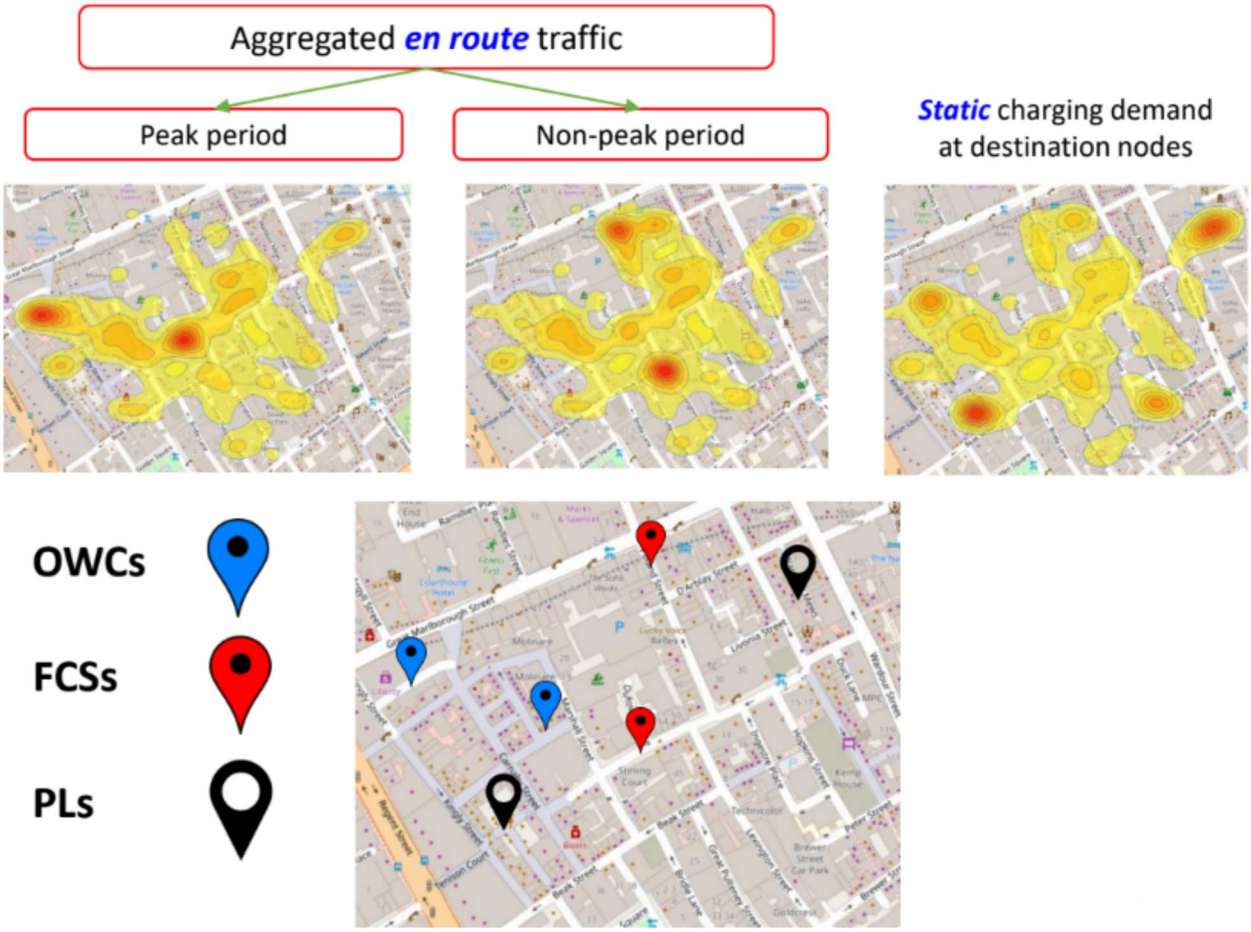

Fig. 8. Covering *en route* and *static* EV charging demand.

Additional presumptions include the existence of C independent, similar chargers at every location, first-come, first-served EV service, and an adequate number of waiting areas. Then, each charging facility in the system is modeled using the $M/M/C$ queuing theory[38], where the mean arrival rate during rush hours is dependent on recorded traffic flows, EV penetration rate, and facility type. The mean arrival rate can be estimated as follows:

$$\lambda_a^1 = \max\{P.(1-\sigma).\beta.f_a(t), \quad \forall t \in T\} \quad \forall a \in \mathrm{O}, \tag{15a}$$

$$\lambda_k^2 = \max\{P.(1-\sigma).(1-\beta).f_k(t), \quad \forall t \in T\} \quad \forall k \in \mathrm{F}, \tag{15b}$$

$$\lambda_d^3 = \max\{P.\sigma.f_d(t), \quad \forall t \in T\} \quad \forall d \in \mathrm{P}, \tag{15c}$$

where the mean arrival rate during rush hour for OWC at link a, FCS at node k, and PL at destination node d, respectively, is denoted by the variables λ_a^1, λ_k^2, and λ_d^3. The P symbol represents the EV penetration rate. The percentage of EV drivers that choose to charge at PL when they get to their travel destinations is indicated by σ. β is the proportion of EV in OWCs that have wireless power transfer capabilities. The time-varying traffic volumes that are handled by OWC, FCS, and PL, respectively, are represented as $f_a(t)$, $f_k(t)$, and $f_d(t)$.

The minimal number of chargers allotted at a charging facility must be greater than the traffic intensity ρ and meet the following inequality, according to the fundamental stability criteria of the $M/M/C$ queuing model:

$$C > \rho = \frac{\lambda}{\mu}. \tag{16}$$

Framing the size of a charging facility as an optimization issue is necessary. The goal of the objective function in (17a) is to reduce the quantity of chargers needed at a charging station. The purpose of constraint (17b) is to ensure that the maximum permissible waiting time W^{Th}, which is predetermined, is not exceeded by the predicted waiting time $\overline{W}$ for consumers at the charging facility. To maintain system stability, constraint (17c) also makes sure that the number of chargers chosen is an integer, indicated by $\lceil c \rceil$, and that it surpasses the traffic intensity.

$$\min_{C} \quad C, \tag{17a}$$

$$s.t. \ \overline{W} \leq W^{Th}, \tag{17b}$$

$$\lceil c \rceil > \rho, \tag{17c}$$

where the expected waiting time can be calculated using the delay probability or Erlang-C formula $\mathbb{P}(W > 0)$, as follows:

$$\overline{W} = \frac{\mathbb{P}(W > 0)}{\mu(C-\rho)}, \tag{18a}$$

$$\mathbb{P}(W > 0) = \frac{\dfrac{C\rho^C}{C!(C-\rho)}}{\displaystyle\sum_{k=0}^{C-1} \frac{\rho^k}{k!} + \frac{C\rho^C}{C!(C-\rho)}}. \tag{18b}$$

The search algorithm that tackles the sizing difficulty is akin to the one described in Ref.[39]. Using this method, the starting number of charges is set as an integer greater than the traffic intensity, and C is applied iteratively until all requirements are satisfied. The mean charging times correspond directly to the typical average durations observed at different types of EV charging facilities. These are assumed as follows: three hours for PL, corresponding to $\mu = 0.3$; twenty minutes for FCS, corresponding to $\mu = 3$; and ten minutes for OWCs, corresponding to $\mu = 6$. The maximum wait periods at OWC, FCS, and PL are five, fifteen, and sixty minutes, respectively. In the system, EV penetration is $P = 20\%$, with PL preferred by 60% of customers and OWC charging available for 30% of EV[40,41].

EV charging demand estimation

The MG load demand includes the assessment of each facility's EV charging needs. The average number of chargers in use at time t at charging facility n, represented as $\mathbb{E}[B_{t,n}^{Ch}]$, is the basis for this estimation. Furthermore, the charging efficiency $\eta_n^{Ch} = 90\%$ and charging power P_n^{Ch} of chargers at facility n are taken into account. In every facility, homogeneous chargers with the same charging power are expected. PL chargers are assumed to use AC Level 3 chargers with a power rating of 43.5 kW. While current PL installations typically feature AC Level 2 chargers with power levels up to 22 kW, the use of AC Level 3 in this study reflects a forward-looking scenario. This assumption anticipates the increasing adoption of higher-capacity chargers in residential and workplace settings as EV penetration and charging demands grow. FCS have DC Level 2 chargers with a 90 kW charging capacity[36]. While standards for wireless chargers are still being worked out, OWC is expected to use 22 kW wireless charging panels[41]. The following formula may be used to determine the power demand $P_{t,n}^{EV}$ of the charging facility n during time slot t:

$$\mathbb{E}[B_{t,n}^{Ch}] = \frac{\lambda_{t,n}}{\mu_n}, \tag{19a}$$

$$P_{t,n}^{EV} = \eta_n^{Ch} P_n^{Ch} \mathbb{E}[B_{t,n}^{Ch}]. \tag{19b}$$

ES system allocation

The goal at this step is to allocate BSS in the grid-connected mode of the MG in the best possible way to improve voltage stability ($f(x)_2$), support the voltage profile ($f(x)_2$), and reduce total power losses ($f(x)_1$). The voltage profile and voltage stability index are critical aspects of power system management. The voltage profile reflects the variation of voltage levels across different buses, playing a vital role in ensuring the quality of the electrical supply[42]. A stable voltage profile minimizes the risk of damage to equipment and enhances the efficiency of power transmission by reducing losses. Conversely, the voltage stability index measures a system's ability to maintain steady voltages during normal operations and after disturbances. This index serves as a predictive tool, helping to identify potential instability issues before they escalate into failures or blackouts[43]. Together, these metrics are essential for optimizing power system reliability, planning, and operational decision-making, ultimately contributing to a more resilient and efficient electrical infrastructure[44].

Employing a multi-objective optimization technique using SOS will facilitate the exploration of various configurations. This process should consider operational constraints like voltage limits and power flow restrictions. After running simulations, the analysis should reveal the optimal locations and sizes of ESS installations that meet the multi-objective criteria such as improving voltage levels, enhancing stability metrics, and reducing power losses. Equation (20) describes the multi-objective function, considering the same weight for the elements:

$$obj = M_1 \times f(x)_1 + M_2 \times f(x)_2 + M_3 \times f(x)_3, \tag{20}$$

$$f(x)_1 = \min \sum_{t=1}^{T} P_{\mathrm{Los},t}, \quad \forall t, \tag{21}$$

$$f(x)_2 = \max V_{j,t} \quad \forall j, \forall t, \tag{22}$$

$$f(x)_3 = \max VSI_{j,t} \quad \forall j, \forall t, \tag{23}$$

where $P_{Los,t}$ expresses the power losses of the grid-connected MG at time *t*. $V_{j,t}$ and $VSI_{j,t}$ express the voltage bus and voltage stability index of bus number *j* at time *t*.

M_1, M_2, and M_3 represent the weighting factors for reducing total power losses, supporting the voltage profile, and improving voltage stability respectively. These factors are instrumental in balancing the trade-offs between minimizing power losses, maintaining the voltage profile, and enhancing voltage stability. The optimization seeks to minimize $P_{\mathrm{Los},t}$, which is influenced by the V_j and the resistance of the transmission lines (R_j). Each node's load and generation capabilities affect the overall power loss. Equation (24) is used to calculate the value of total power losses of the MG.

$$P_{\mathrm{loss}} = \sum_{j=1}^{mb} \frac{V_j^2}{R_j}. \tag{24}$$

The optimization includes maximizing the voltage $V_{j,t}$ at each node *j* at time *t*. The voltage levels are critical for ensuring that power is delivered effectively. Each node's voltage is influenced by its load demand, generation sources, and the configuration of the network. A higher voltage level typically indicates better performance and reliability. $V_{j,t}$ can be calculated by applying power flow equations: For each bus *j*, use the power flow Eq. (25).

$$V_j = \frac{1}{Y_{jj}} \left(P_j - jQ_j + \sum_{i \neq j} V_j Y_{ji}^* \right). \tag{25}$$

Here, V_j is the voltage in bus *j*, Y_{jj} is the diagonal element of the admittance matrix *Y* for bus *j*, P_j and Q_j are the injected real and reactive powers, respectively, and Y_{ji} is the admittance matrix element.

Maximizing the VSI at each node *j* at time *t* assesses the node's ability to maintain stable voltage under varying conditions. A higher VSI signifies better voltage stability, which is vital for preventing voltage collapse during peak loads or disturbances. Equation (26) may be used to assess the voltage stability of MG, with low-index value buses being the most vulnerable to voltage failure[11].

$$VSI(i) = V_s^4 - 4(PX - Q\mathscr{R})^2 - 4V_s^2(P\mathscr{R} + QX), \tag{26}$$

where the line reactance is *X*, the line resistance is $\mathscr{R}$, the line transferred reactive power is *Q*, the line transmitted active power is *P*, and the transmitting voltage magnitude is V_s.

Balance of electrical demand
The whole electrical demand and power charged to the BSS considering the total demand scenarios *nsl* and their probability Φ_{sl-t}, as specified in Eq.(27), must be equal to the total generated power from the main grid, the power discharge from the BSS, and all RESs considering the various scenarios for PV output power *nsp* and their chances $\Phi_{i,sp-t}$ and the various scenarios for WT output power *nsw* and their chances $\Phi_{i,sw-t}$.

$$\sum_{i=1}^{npv}\sum_{sp=1}^{nsp} P_{i,sp-t}^{pv}\Phi_{i,sp-t} + \sum_{i=1}^{nwt}\sum_{sw=1}^{nsw} P_{i,sw-t}^{wt}\Phi_{i,sw-t} + P_t^{grid} + P_{dis-t}^{BSS} = \sum_{sl=1}^{nsl} P_{sl-t}^{l}\Phi_{sl-t} + P_{ch-t}^{BSS} + P_t^{EVCh} \quad \forall t, \tag{27}$$

where $P_{i,sp-t}^{pv}$ is the output power of PV unit *i*. *npv* represents total number of PV units. $P_{i,sw-t}^{wt}$ is the output power of wt unit *i*. *nwt* represents total number of WT units. P_t^{grid} is the power of main grid.P_{dis-t}^{BSS} and P_{ch-t}^{BSS} represents the discharge and charge power of BSS.P_t^{EVCh} represents the load of EV on the MG.

Limits of RESs
The output power of each WT ($P_{i,sw-t}^{wt}$) is constrained by its minimum ($P_{i,sw-t,min}^{wt}$) and maximum power ($P_{i,sw-t,max}^{wt}$) as shown in Eq. (28). The output power of PV ($P_{i,sp-t}^{pv}$)is constrained by its minimum ($P_{i,sp-t,min}^{pv}$)and maximum power ($P_{i,sp-t,max}^{pv}$) as shown in Eq. (29).

$$P_{i-t,min}^{wt} \le P_{i-t}^{wt} \le P_{i-t,max}^{wt} \forall t, \tag{28}$$

$$P_{i-t,min}^{pv} \le P_{i-t}^{pv} \le P_{i-t,max}^{pv} \forall t. \tag{29}$$

Voltage boundary
Bus voltage (V_{j-t}) in each bus *j* at any time *t* should be regarded as falling within minimum (V_{j-t}^{min}) and maximum (V_{j-t}^{max}) allowed range as shown in Eq. (30):

$$V_{j-t}^{min} \leqslant V_{j-t} \leqslant V_{j-t}^{max} \quad \forall j, \forall t. \tag{30}$$

Energy storage limits
The charging (P_{ch-t}^{BSS}) and discharging (P_{dis-t}^{BSS}) power of the BSS must not exceed the maximum value of charge ($P_{ch-t,max}^{BSS}$) and discharge ($P_{dis-t,max}^{BSS}$) power.

$$P_{ch-t}^{BSS} \le P_{ch-t,max}^{BSS}, \forall t, \tag{31}$$

$$P_{dis-t}^{BSS} \le P_{dis-t,max}^{BSS}, \forall t. \tag{32}$$

The state of charge of the BSS at any time (S_t^{BSS}) must be regarded as falling within minimum ($S_{t,min}^{BSS}$) and maximum ($S_{t,max}^{BSS}$) values as shown in Eq. (33)

$$S_{t,min}^{BSS} \le S_t^{BSS} \le S_{t,max}^{BSS} \forall t. \tag{33}$$

Additionally, as shown by Eq. (34), the percentage of discharged power should match the percentage of charged power considering the BSS efficiency (η).

$$sum_{t=1}^{24} P_{dis-t}^{BSS} = sum_{t=1}^{24} P_{ch-t}^{BSS} * \eta. \tag{34}$$

The present (S_t^{BSS})is related to the previous S_{t-1} and the power charge and discharge of BSS at present *t* as given in Eq. (35). The initial SoC S_{in}^{BSS} must be considered at the first time $(t = 1)$ as given in Eq. (35).

$$S_t^{BSS} = \begin{cases} S_{in}^{BSS} + \eta_{\text{ bat}} P_{ch-t}^{BSS}\Delta t - \Delta t P_{dis-t}^{BSS} \ t = 1 \\ S_{t-1} + \eta_{bat} P_{ch-t}^{BSS}\Delta t - \Delta t P_{dis-t}^{BSS} \ \forall t \ge 2, t \in T \end{cases} \tag{35}$$

Economic aspect
The comprehensive operational cost (*CMG*) of the MG under review throughout the day includes expenses for purchasing kilowatt-hours from the main grid, as well as costs associated with electricity generation from PV ($C_{t,i}^{PV}$) sources and WT ($C_{t,i}^{WT}$). Additionally, the total cost of BSS per day (*CBSS*) ($/day) through the lifetime of the project (*PLF*).This is detailed in Eq. (36).

$$\begin{aligned} CMG = \sum_{t=1}^{T} \Bigg[\left(Price_t^{\text{grid}} \times \text{P}_t^{\text{grid}}\right) + \sum_{i=1}^{npv}\sum_{sp=1}^{nsp} \left(\Phi_{i,sp-t} \times C_{t,i}^{PV} \times \text{P}_{i,sp-t}^{PV}\right) \\ + \sum_{i=1}^{nwt}\sum_{sw=1}^{nsw} \left(\Phi_{i,sw-t} \times C_{t,i}^{WT} \times \text{P}_{i,sw-t}^{WT}\right) \Bigg] + CBSS \end{aligned} \tag{36}$$

The *CBSS* ($/day) over *PL* includes the capital cost of BSS (*BSSI*) as well as the expenses associated with the number of replacements of the BSS (REP_S). *BSSI* is associated with the battery's power (P_{BSS}) and energy

(E_{BSS}) capabilities within the storage system as shown in Eq.(37). C^{PBSS} denotes the power rating cost of the BSS ($/kW), representing the expense associated with the power output capability of the BSS. E_{BSS} represents the capacity rating cost of the BSS ($/kWh), signifying the cost related to the energy storage capacity of the BSS, indicating the total amount of energy that can be stored and discharged over a specific period.

$$BSSI = \left(C^{PBSS} \times P_{BSS}\right) + \left(C^{EBSS} \times E_{BSS}\right). \tag{37}$$

To calculate the replacement number of the BSS, the process begins by determining the number of cycles completed by the BSS (S_C). This is achieved using Eqs. (38) and (39), which help assess the overall lifetime of the BSS. Once S_C is established, Eq. (40) is utilized to compute the lifetime of the BSS F_S. This calculation relies on both the total life cycles of the BSS($S_{\text{Lifecycle}}$, and S_C.

$$N_S(t,d) = \left(n_{S(t)} - n_{S(t-1)}\right) n_{S(t)}, \quad \forall t \in T, \forall d \in D, \tag{38}$$

$$S_C = \sum_{d=1}^{D} \sum_{h=1}^{H} N_S(h,d), \tag{39}$$

$$F_S = \frac{S_{\text{Lifecycle}}}{S_C}, \tag{40}$$

where $N_S(t,d)$ denotes the BSS cycles, while *D* indicates the total number of operating days per year, which is set in this paper to 365. The BSS operates in two modes: charging and discharging. The variable $n_{S(t)}$ represents the status of the BSS for each hour during operating days in the year, where $n_{S(t)}$ is equal to 1 when the BSS is charging and 0 when the BSS is discharging. Consequently, the replacement number for BSS (REP_S) over the (*PLF*) is defined by Eq. (41)

$$REP_S = \frac{PL}{F_S}. \tag{41}$$

Therefore, *CBSS* ($/day) can be determined by using Eq. (42), which is a function of the interest rate *i*.

$$CBSS = \frac{1}{D \cdot PL}\left(\frac{i(1+i)^{PLF}}{(1+i)^{PLF}-1} \cdot BSSI \cdot REP_S\right). \tag{42}$$

To conduct a thorough analysis of the operational costs of the MG, the selection of a BSS plays a pivotal role. To comprehensively analyze the operational expenses of the MG, the choice of a BSS is crucial. Given the diverse traits of BSSs, our model evaluates five distinct BSS technologies to pinpoint the most fitting one for integration into the MG. The primary aim is to reduce the overall operational costs of the system effectively. In this study, five distinct battery types are employed: Sodium-Sulfur (NaS), Sodium-Nickel Chloride (Na-NiCl2), Lead-Acid (Pb-A), Nickel-Cadmium (NiCd), and Lithium-Ion (Li-ion). The details regarding the cost considerations, efficiency metrics, and lifecycle particulars of the five batteries employed in this evaluation are presented in Table 4[14,21,22] . The bids for PV energy stand at 0.12 dollars per kilowatt-hour ($0.12/kWh), while the bids for WT energy are 0.08 dollars per kilowatt-hour ($0.08/kWh). Figure 9 shows the market price of the main grid during the day[24] .

Symbiotic organisms search

SOS is a nature-inspired metaheuristic optimization algorithm that is based on the symbiotic relationship between organisms in nature that may be used to solve optimization problems, both limited and unconstrained[45]. The mutualism phase: During this stage, The organism *j* and another randomly picked ecosystem organism *h* communicate with each other through the vector Y_i of the creature *j* and the vector Y_h of the creature *h* (where *h* doesn't equal *j*).A mutualistic connection is created by various creatures interacting, and this relationship enhances each organism's unique functional value within the ecosystem. As a result, Mutual Vectors (Y_{Mutual}) and Benefit Factors (BF^1 and BF^2) control the development of new species. The relationship between organisms Y_i and Y_h is shown by the mutual vector, which is the average of the two organisms (Eq. (45)). The benefit factors are determined at random with equal probability using a heuristic process, either 1 or 2 (Eqs. (46) and (47))). Thus, the benefit factors denote two conditions in which organisms Y_j and Y_h, respectively, gain entirely or

ESS	NaS	Na-NiCl2	Pb-A	NiCd	Li-ion
C^{PBSS} ($/kW)	350	200	300	500	900
C^{EBSS} ($/kWh)	300	100	400	400	600
Lifetime (years)	15	15	15	20	15
η (%)	90	90	80	85	90
$S_{Lifecycle}$	4000	4000	2000	3000	5500

Table 4. Economic aspects of different batteries.

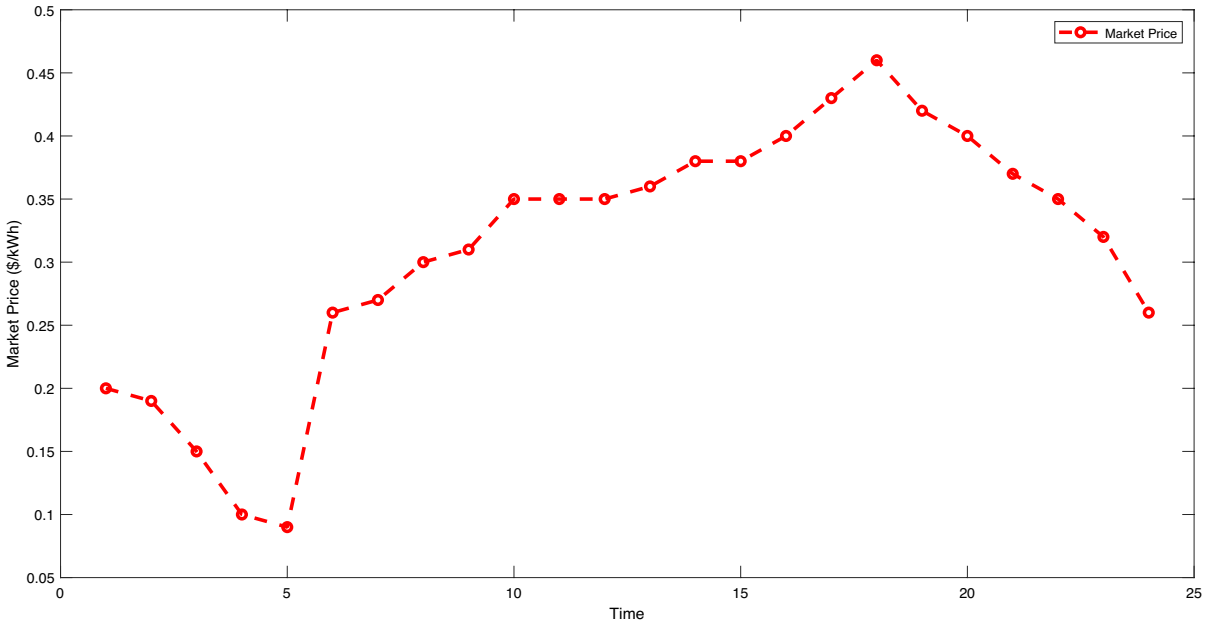

Fig. 9. Hourly values of the market price during a typical day.

partially from the contact. The best organism (Y_{best}) in the ecosystem is defined as the one having the highest functional value. The best creature interacts with organisms Y_j and Y_h during this period as well. As a result, this stage maintains a healthy equilibrium between search space exploration and exploitation. Only when the organism's new functional value ($F(Y_j')$ or (Y_h')) outperforms the current one is it updated. The new solutions' mathematical formulations are provided in Eqs. (43) and (44))).

$$Y_j' = Y_J + \text{rand} * \left(Y_{\text{best}} - Y_{\text{Mutual}} * BF^1\right), \tag{43}$$

$$Y_h' = Y_h + \text{rand} * \left(Y_{\text{best}} - Y_{\text{Mutual}} * BF^2\right), \tag{44}$$

$$Y_{\text{Mutual}} = \frac{X_j + X_h}{2}, \tag{45}$$

$$BF^1 = 1 + \text{round}[\text{ rand }], \tag{46}$$

$$BF^{=}1 + \text{round}[\text{ rand }], \tag{47}$$

where $h \in (1, 2, \ldots, n)$ is a randomly selected population, $i = 1, 2, \ldots, n$, $k \neq i$; The commensalism phase: Choosing a random creature, Y_h, to interact with a certain organism, Y_j, in the environment is known as commensalism. This form of association occurs when one organism gains while the other is neither aided nor hurt. In this case, Y_j benefits from the contact but Y_j is left unchanged. The ecosystem update for Y_j is performed with the following equation:

$$Y_j' \text{ new } = Y_j + \text{rand}(-1, 1) * (Y_{\text{best}} - Y_j). \tag{48}$$

Here, $(Y_{\text{best}} - Y_j)$ represents the advantage that Y_j gets from Y_h. Only when Y_j' new outperforms Y_j in terms of fitness value does Y_j' new replace Y_j in the ecosystem; otherwise, Y_j stays in the ecosystem and Y_j' new is eliminated.

The parasitism phase: The term "parasite" refers to the beneficial organism in this case. An organism Y_j is selected and multiplied within an environment to create a vector for parasites. Concurrently, a different organism Y_h is chosen at random from the environment to act as the host of this parasite vector. The parasite vector replaces Y_h if its fitness value is greater than Y_h's. If Y_h, on the other hand, shows a higher fitness value, it stays the fittest and the parasite vector dies.

The SOS optimization method is selected for several key reasons. Firstly, inspired by natural symbiotic relationships, SOS can efficiently navigate intricate solution spaces by leveraging interactions between different organisms, potentially offering innovative solutions beyond traditional methods. Secondly, SOS's inherent flexibility and adaptability make it well-suited for various problem types, especially in multi-objective optimization scenarios. It effectively balances objectives like maximizing voltage stability and minimizing power losses. Additionally, the algorithm has demonstrated robust performance across diverse optimization challenges, often outperforming other techniques and ensuring reliable solutions in complex power system contexts. SOS excels at striking a balance between exploration and exploitation, crucial in multi-objective optimization,

where navigating trade-offs among conflicting objectives is paramount. Moreover, its ease of implementation and customization compared to more complex algorithms makes it a practical choice for tailoring solutions to specific research needs. Lastly, SOS's proficiency in handling nonlinear and complex optimization landscapes, commonly encountered in power system issues, enhances its usefulness in addressing problems with non-linear variable relationships[45,46].

Simulation results with analytical discussion

Location planning results of electric vehicle charging stations

To validate the model's effectiveness and showcase its practicality, we opted for a widely recognized RTN known as the Nguyen-Dupuis network[15]. The geographical alignment of RTN nodes/links with the MG buses ensures that each RTN node or link is serviced by an electric bus from the MG. The node number in RTN is assumed to match the MG bus number. A simulation of time-varying traffic flows involving 1500 EV was conducted using the SUMO road traffic simulator[47].

As depicted in Fig. 10, the proposed solution strategically places FCS at nodes {2, 6, 10, and 11} and OWC at link {4} that is corresponding to node {9} to address 50.49% of the overall en route charging demand within the network. Additionally, a PL is deployed at node {3}, effectively covering 55.56% of the total static charging demand. The respective number of chargers in the FCS is {7, 4, 4, 5}, while 44 slow chargers are allocated to the PL and 2 chargers in OWC. The entire EV charging infrastructure siting and sizing model is implemented within a Python 3.10 environment.

The increased demand for electricity needed to charge EV might hinder the MG's ability to operate efficiently. Figure 11 displays the voltage of each bus in the MG at certain hours after considering the demand of the EV charging stations. Notably, during hours 1 to 5, the voltage at buses 9 to 18 and 29 to 33 exhibited an increase over the highest acceptable limit (1.05 P.U.). Additionally, during hour 6, the voltage at buses 16 to 18 and 31 to 33 exhibited an increase also over the highest acceptable limit (1.05 P.U.) Conversely, the voltage profile curve revealed instances of voltage decline below the specified boundary (0.95 P.U.) at various hours and buses within the MG. Specifically, during hours 11 to 12 and 14 to 17, buses 10 to 18 experienced a notable drop in voltage magnitude. Furthermore, during hours 10 and 18, buses 13 to 18 exhibited a decline in voltage below the specified boundary. Additionally, in hour 9, buses 10 to 18 experienced a decline in voltage below the specified boundary. Furthermore, in hour 13, buses 11 to 18 experienced a notable drop in voltage magnitude below the specified boundary. As one can see, the demand for EV charging stations negatively affected the voltage profile within the considered MG.

Location planning results of energy storage system

Table 5 presents the optimal allocation of BSS to improve MG performance after adding the power demand of EV charging stations by lowering overall power losses, supporting the voltage profile, and boosting voltage stability. The SOS algorithm efficiently reached the optimal BSS allocation through 1000 iterations in just 12.64

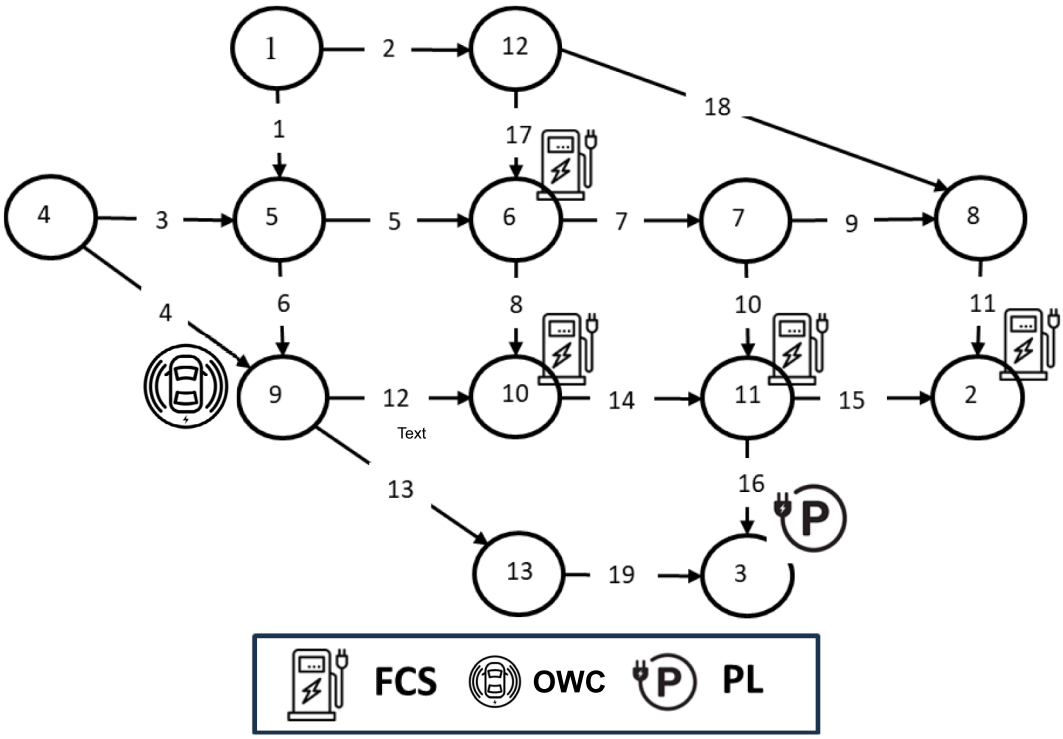

Fig. 10. Allocated EV charging infrastructure.

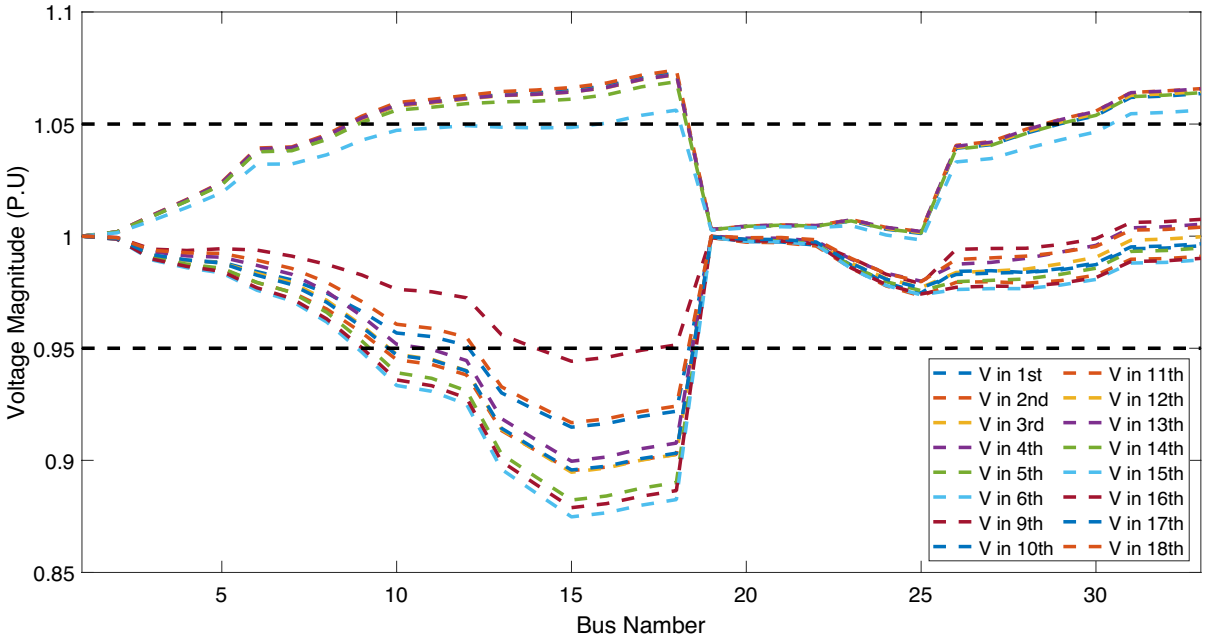

Fig. 11. Hourly Voltage profile of 33 buses before adding BSS.

Case	BSS location	BSS (P) (MW)	BSS (E) (MWh)	Iteration	Time (s)
BSS	15	1.943	11.627	1000	12.64

Table 5. Optimal location and capacity of the BSS.

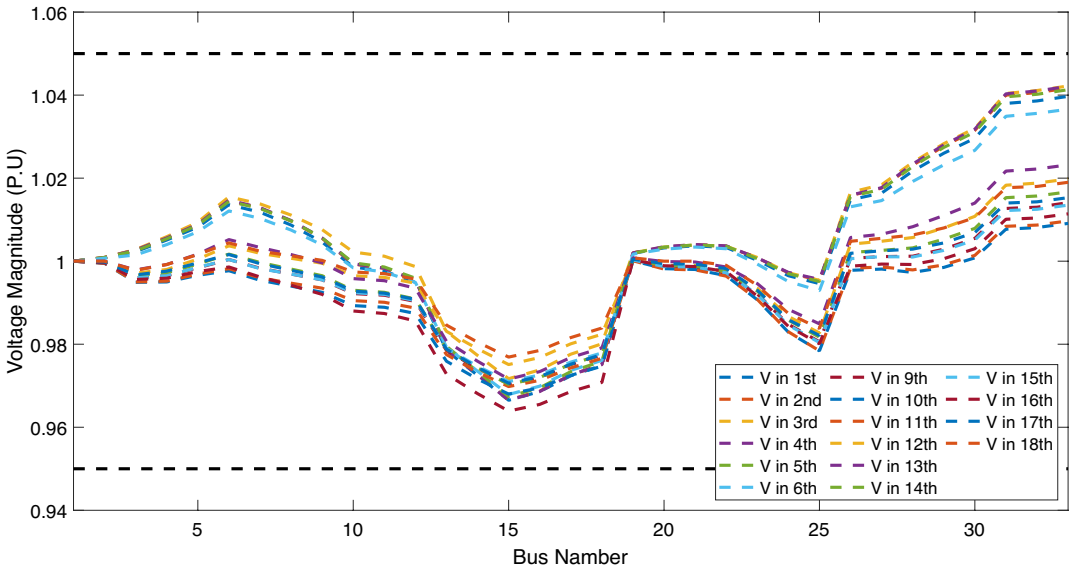

Fig. 12. Hourly voltage profile for 33 buses after installing BSS.

seconds, showcasing its ability to converge effectively despite typically having moderate to high optimization times influenced by factors such as population size, convergence criteria, and problem complexity.

Figure 12 shows the voltage profile of the MG after adding BSS in the hours that voltage problems occurred before inserting BSS. It is clear that from the obtained results, BSS has a significant influence on regulating and stabilizing the voltage within the desired range, ensuring reliable and efficient operation of the MG.

It is noted that before adding the BSS, the minimum VSI value is 0.5848 at hour 15 on bus 15, while after adding the BSS, the minimum VSI value is 0.8631 at hour 16 on bus 16. This indicates an increase in the VSI

after adding the BSS compared to the VSI before adding the BSS. In this aspect, the BSS has a positive impact on improving the VSI of the studied MG.

Figure 13 demonstrates the total power losses of the considered MG through each hour before and after adding BSS. Before integrating the BSS, the MG's daily power losses were 5616.7 kW, considering the electrical demand of the EV charging plants. However, after integrating the BSS, the total daily power losses were reduced to 3744.1 kW. This indicates a decrease in power losses by 1872.6 kW. Therefore, the daily percentage decrease in the MG's overall power losses is approximately 33.34 %. This indicates a significant reduction in power losses after integrating the BSS, considering the electrical demand of the EV charging plants. In this regard, the BSS positively affects the MG's overall power losses.

It is interesting to note from Fig. 13 that the maximum transformer loading that occurs in case the MG runs without BSS is 6.6385 MW at hour 16, and when the MG operates with BSS the maximum value of transformer loading decreases to 5.3385 MW at hour 16. As a result, the BSS is crucial in lowering the transformer load rate by a percentage of 19.5%.

Figure 14 presents the hourly SoC, charging, and discharging power of BSS. The chosen charging and discharging schedule supports the overall performance of the MG.

In this study, the uncertainties related to electrical demand and RESs were addressed by generating multiple scenarios, each assigned a probability, through a roulette wheel mechanism. To manage computational complexity, we employed the fast-forward scenario reduction method, which effectively decreases the number of scenarios while preserving the most representative ones. For the final output values shown in Table 6, a weighted average of the scenario results was computed, with each scenario's contribution proportional to its probability of occurrence. Table 6 shows the results obtained when considering uncertainty parameters such as load variations and fluctuations of RESs and without considering uncertainty parameters.It is clear from taking these uncertainty metrics into account that load changes and variations in the production of renewable energy have a big impact on a lot of different elements of the MG's performance and operation. Understanding these effects allows for better planning, optimization, and risk assessment in MG design and operation.

The results show that over the 15-year duration of the project, the battery system will complete approximately 5,475 charge and discharge cycles. Based on the characteristics of each battery type, sodium-sulfur batteries will require one replacement due to their limited cycle life. Similarly, sodium-nickel chloride batteries will also need one replacement during the project period. Lead-acid batteries will need to be replaced twice, while nickel-cadmium batteries will require one replacement. In contrast, lithium-ion batteries are expected to last the entire project duration without needing replacement, owing to their longer cycle life compared to other technologies. Table 7 presents the economic results of five ESS: NaS, Na-NiCl2, Pb-A, NiCd, and Li-ion. It outlines key financial metrics, including initial investment costs, the number of replacements needed over the project's lifespan, total investment, daily investment costs, total operational costs of the MG, and percentage savings. Na-NiCl2 emerges as the most cost-effective option, with the lowest initial investment and daily costs, while Lithium-Ion, despite its higher upfront cost, requires no replacements, resulting in significant long-term savings. Pb-A, on the other hand, incur the highest total investment and operational costs, highlighting the importance of considering both initial and ongoing expenses when evaluating energy storage technologies. Table 7 shows the savings achieved with each battery system compared to a baseline scenario. Na-NiCl2 offers the highest savings at 6.99%, while Pb-A provides the lowest at 0.94%.

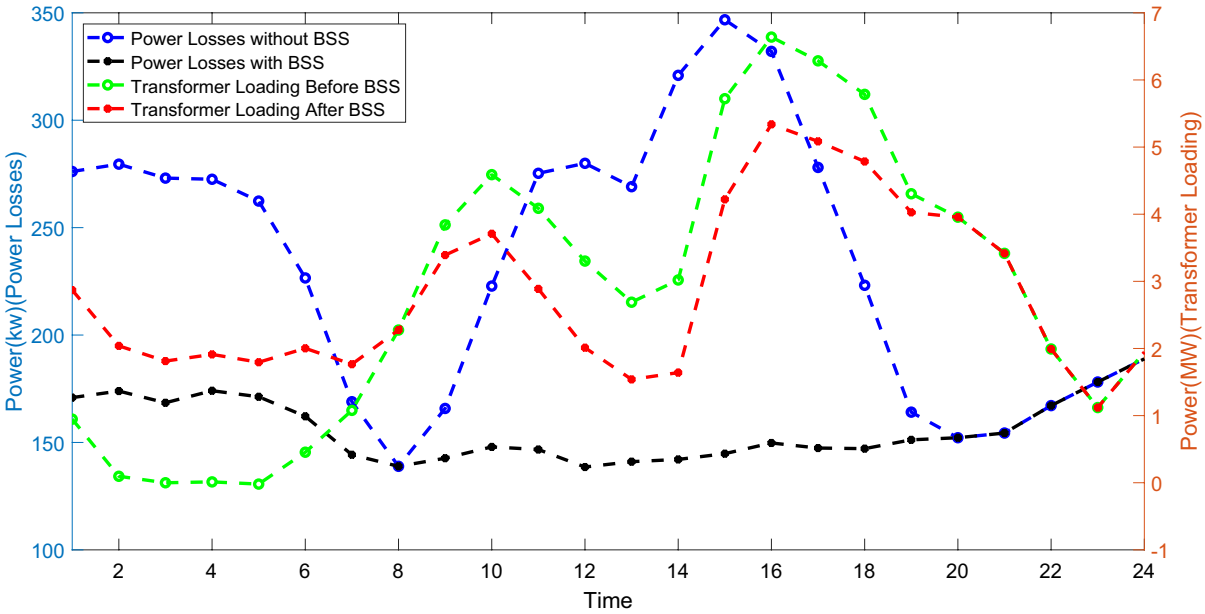

Fig. 13. MG hourly power losses and transformer load rate with/without BSS considering EV charging station loads.

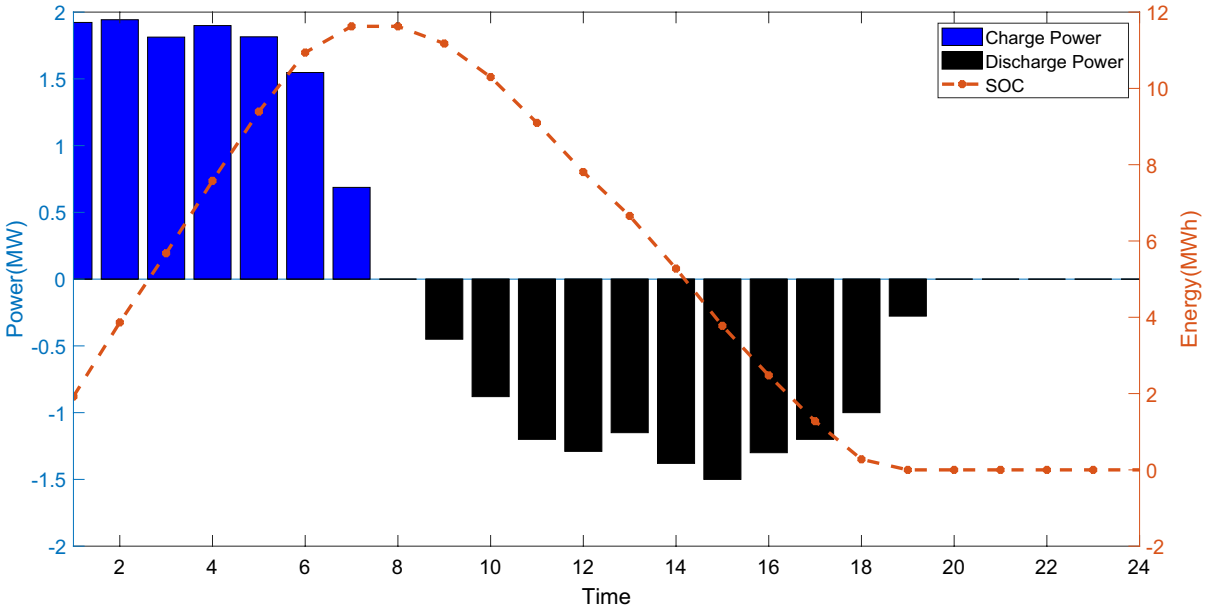

Fig. 14. BSS SoC and hourly BSS's charging/discharging power.

Item	With uncertainty	Without uncertainty
BSS location	15	15
BSS power (MW)	1.943	2.1476
BSS energy (MWh)	11.627	11.325
Highest value of transformer loading before BSS (MW)	6.63 MW at *t* 16	6.52 MW at *t* 16
Highest value of transformer loading after BSS (MW)	5.33 MW at *t* 16	5.42 MW at *t* 16
MG's daily losses before BSS (kW)	5616.7	5534.2
MG's daily losses after BSS (kW)	3744.1	3748.2

Table 6. The obtained outcomes with/without considering uncertainty parameters.

ESS	NaS	Na-NiCl2	Pb-A	NiCd	Li-ion
CMG without BSS($)	29168.4	29168.4	29168.4	29168.4	29168.4
BSSI ($)	4168150	1551300	3973850	5622300	8724900
REP_S	1	1	2	1	0
CBSS including REP_S ($)	8336300	3102600	11921550	11244600	8724900
CBSS including REP_S/day($)	1522.6	566.6	2177.4	2053.8	1593.5
CMG/day ($)	28217.9	27262.1	28897.3	28762.1	28288.9
Saving %	3.47	6.99	0.94	1.41	3.11

Table 7. Economic results achieved by SOS.

Conclusions

In this study, an integrated planning framework based on the ST-FCLM is proposed to strategically locate EV charging facilities within the transportation network, ensuring the optimal allocation of EV charging stations. The sizes of these designated charging stations are determined by applying queuing theory. Following this, the impact of EV charging station loads on the performance of the MG, including voltage profile, power loss, transformer loading, and the voltage stability index, is thoroughly analyzed.To further enhance MG performance, a novel mathematical model is introduced to allocate the BSS within the MG, addressing the uncertainties of electrical demand and the intermittency of RESs. This model aims to reduce overall power losses, improve voltage profiles, and increase the voltage stability index. The results demonstrate that fluctuations in renewable energy generation and changes in load significantly affect various aspects of MG operation and performance. A better understanding of these effects enables more informed planning, optimization, and risk assessment in MG design and operation. A key contribution of this research lies in examining the impact of optimally allocating BSS to

mitigate the challenges posed by the integration of EV in medium-sized MGs. The obtained results indicate that the integration of BSS significantly enhanced voltage stability, with the minimum VSI rising from 0.5848 to 0.8631. Furthermore, power losses were reduced by 33.34%, transformer loading decreased by 19.5%, and overall economic efficiency improved. Notably, Na-NiCl2 provided the highest savings at 6.99%, highlighting substantial technical and financial advantages for the MG. While applying SOS to address these optimization challenges, it is recognized that the complexity of the algorithm can impact its effectiveness and scalability in more intricate optimization scenarios. The inherent complexity of simulating symbiotic relationships in optimization problems introduces additional challenges, necessitating a deep understanding of optimization techniques. Although techno-environmental considerations, such as reducing MG emissions while optimizing operating costs, were not included in the scope of this study, these aspects will be explored in future research.

Data availability

All data generated or analyzed during this study are included in this published article. Also, the datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Received: 14 May 2025; Accepted: 15 July 2025
Published online: 25 July 2025

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Acknowledgements
The authors gratefully acknowledge Qassim University, represented by the Deanship of Scientific Research, on the financial support for this research under the number (2023-SDG-1-BSRC35415) during the academic year 1445 AH / 2023 AD.

Author contributions
T. A., M.H.M. and A. A designed the problem under study;T. A., M.H.M.,and A. A. performed the simulations and obtained the results. T. A. analyzed the obtained results. T. A., M.H.M.,and A. A. wrote the paper, which was further reviewed by T. A. All authors reviewed the manuscript

Declarations

Competing interests
The authors declare no competing interests.

Additional information
Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-025-12079-3.

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